

Problem K. Diameter of a Tree

OS Linux

Given an array of unique elements, construct a Binary Search Tree and find the diameter of the tree. Diameter is defined as the number of nodes on the longest path between 2 nodes of the tree.

Input Format

First line of input contains T - number of test cases. Its followed by 2T lines. First line of each test case contains N - number of nodes in the BST. The next line contains N unique integers - value of the nodes.

Constraints

1 <= T <= 1000
 1 <= N <= 5000
 0 <= ar[i] <= 10000

Output Format

For each test case, print the diameter of the Binary Search Tree, separated by newline.

Input	Output
3 5 1 2 3 4 5 5 2 4 3 1 5 7 4 5 15 0 1 7 17	5 4 6

Explanation 0

Self Explanatory